

# A Compelling Image: The Tower of Babel and the Proliferation of Programming Languages During the 1960s

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*In 1961 and in 1969, two towers of Babel of programming languages were published. The first one appeared on the cover of Communications of the ACM. The second one appeared on Jean E. Sammet's Programming Languages: History and Fundamentals. These two towers have come to symbolize the fragmentation that plagued the development of programming as a result of the multiplicity of notations. I will argue that contrary to this common view, the tower on the cover of Communications should be understood as a proud display of the research that profoundly transformed computer programming during the late 1950s. The tower of Sammet's book reflected instead the book's goal of bringing order into the field of programming languages. Symbolically, Sammet's book marked the end of a decade that witnessed the emergence of a logico-linguistic approach to the study of programming notations, a decade that also spurred Sammet's fascination with Babel.*

## INTRODUCTION

Particularly during the 1960s, the illustration of a tower of Babel of programming languages became a prominent way of portraying what Nathan Ensmenger has graphically called the “Cambrian explosion” that characterized the development of programming languages.<sup>1</sup> This form of visual appropriation of the myth of Babel dates back to the publication of a now iconic tower of programming languages, techniques, and systems on the cover of the January 1961 issue of *Communications of the ACM*, the flagship journal of the United States-based Association for Computing Machinery (ACM).<sup>2</sup> Eight years later, in 1969, the front endpaper and the book jacket of Jean E. Sammet's seminal volume *Programming Languages: History and Fundamentals*, which

is still the main reference work for all interested in the history of programming languages, also famously portrayed a tower of Babel of the more than hundred languages covered in the book.<sup>3</sup>

Other visual appropriations of the myth of Babel in computer programming appeared in the same decade. In 1964 the booklet of a scientific conference held near Vienna, which addressed the question of how to formally define the meaning of instructions in programming languages, featured the 1563 painting by Pieter Bruegel the Elder, “The (Great) Tower of Babel,” preserved at the Kunsthistorisches Museum in Vienna.<sup>4</sup> A copy of the painting was also on display at the nearby IBM Laboratory Vienna, which cohosted the

<sup>1</sup>Ensmenger (2010), p. 105.

<sup>2</sup>Comm. ACM (1961).

<sup>3</sup>Sammet (1969).

<sup>4</sup>The IFIP Working Conference on Formal Language Description Languages was held in Baden bei Wien, near Vienna, from 14 to 18 September 1964. On the relevance of this conference for the field of formal methods, see Astarte (2019), Ch. 4. A copy of the conference booklet featuring Pieter Bruegel the Elder's “The (Great) Tower of Babel” is to be found at the Bodleian Libraries, Oxford, as part of the papers of Christopher Strachey (Bodleian Libraries, University of Oxford, MS. Eng. misc. b. 287/E. 36). I thank Troy K. Astarte for providing me with the information concerning the location of this conference booklet.

conference.<sup>5</sup> Then in August 1968, the magazine *Computers and Automation* included a piece of computer-generated art by Leslei Mezei and David Payne from the Institute of Computer Science at the University of Toronto consisting of five pictures based on a mathematical distortion of a tower made up of the word "BABEL."<sup>6</sup>

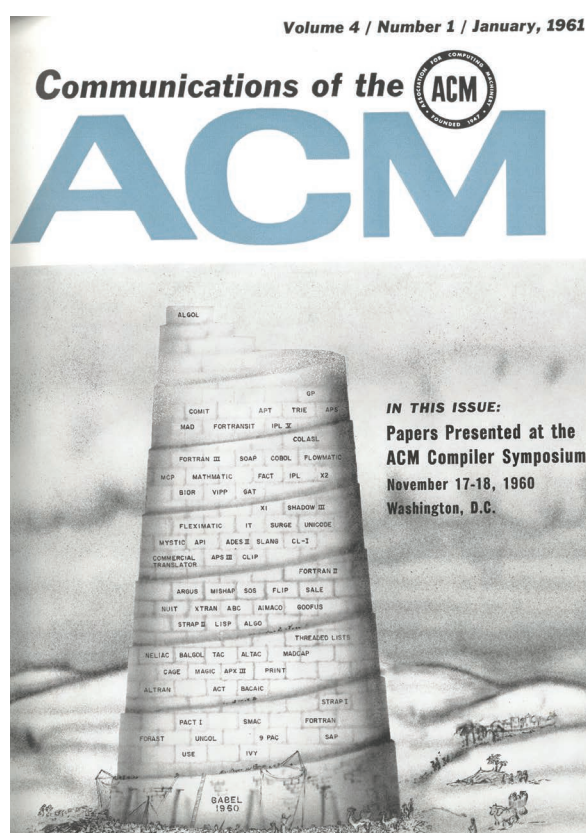
The towers on the covers of *Communications of the ACM* and Sammet's book have acquired a symbolic status as a representation of the endless proliferation of new programming languages. They are therefore the main focus of the present study. These two towers are, however, very different from each other, in terms of both their visual content and style, as well as in terms of the contexts in which they originated and were put to work. Their meaning cannot thus be taken for granted. In this article, I will analyze what these two towers represented, individually and in relation to each other. I will pay particular attention to the ranking of languages in both towers and to the concerns, aspirations, and expectations of those who were responsible, as far as I have been able to ascertain, for these images. And I will show that contrary to what is often assumed, even by historians of computing, these two iconic towers did not primarily seek to convey a messy proliferation of still new programming languages, techniques, and systems.<sup>7</sup>

I will argue that the arrangement of languages, systems, and techniques in the tower of *Communications of the ACM* was an expression of self-confidence, optimism, and positive expectations in the progress in computer programming. Such progress would eventually lead to the development of a language capable of expressing all algorithms in an elegant and

<sup>5</sup>A photograph from the archives of the Technical University Vienna showing the reproduction of Bruegel's painting at the IBM Laboratory Vienna can be found at: M. Wagner, "Heinz Zemanek and the Curious Story of the 'Mailüfterl'". TUWIEN. at. Accessed: Aug. 12, 2024. [Online] Available: <https://informatics.tuwien.ac.at/heinz-zemanek-and-the-curious-story-of-the-mailuefterl/>. I thank one of the reviewers for drawing my attention to this website. The reproduction of Bruegel's painting was displayed first at the TU Vienna in the room where Heinz Zemanek's team developed the first Austrian computer (the "Mailüfterl"). In 1961 Zemanek's group moved together with the reproduction of Bruegel's painting to the newly established IBM Laboratory Vienna (see Astarte (2019), Ch. 5). In his late writings, Zemanek freely interpreted Bruegel's painting as a symbol of the risks inherent in the development of new software systems. See, for example: Zemanek (1992), pp. 103-105, 125-126.

<sup>6</sup>*Computers and Automation* (1968), pp. 12-13.

<sup>7</sup>Historians of computing have tended to follow the 'confusion of tongues' interpretation particularly when referring to the tower on the cover of the *Communications of the ACM*. See: Halvorson (2020), pp. 70-71; Priestley (2010), p. 204; Ensmenger (2010), p.105; Pflüger (2002), p. 137.



**FIGURE 1.** The cover of the January 1961 issue of *Communications of the ACM*.

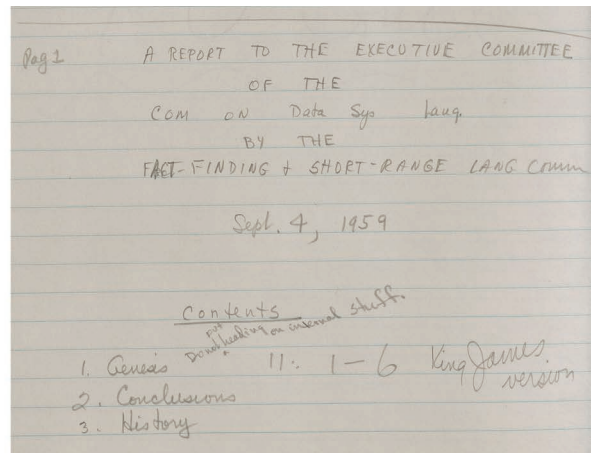
transparent form, or at least, that was the expectation of some of those who were responsible for this iconic cover. Quite different was the tower on the cover of Sammet's book. The alphabetical arrangement of the languages in Sammet's tower contrasted with the optimism and confidence that had motivated the tower of *Communications*, Sammet's source of inspiration. Such alphabetical arrangement conveyed well the purpose of Sammet's book, which sought to bring some order into the complex field of programming languages by providing the reader with a well-ordered framework, with complete information, for the study of the most important languages developed up to 1967. Symbolically, Sammet's book marked the end of a decade in which the idea of a single programming language for all purposes had been seen as the solution to the profusion of programming languages.

When exploring the appropriation of the biblical story in computer programming, it is important to keep in mind the polysemous and open character of the biblical myth. As Biblical studies scholars have shown, the ambiguities, indeterminacies, and silences of the original Hebrew text of Genesis 11:1-9 are constitutive

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1. And the whole earth was of one language, and of one speech.
2. And it came to pass, as they journeyed from the east, that they found a plain in the land of Shinar; and they dwelt there.
3. And they said one to another, Go to, let us make brick, and burn them thoroughly. And they had brick for stone, and slime had they for mortar.
4. And they said, Go to, let us build us a city, and a tower, whose top may reach unto heaven; and let us make a name, lest we be scattered abroad upon the face of the whole earth.
5. And the LORD came down to see the city and the tower, which the children of men builded.
6. And the LORD said, Behold the people is one, and they have all one language; and this they begin to do; and now nothing will be restrained from them, which they have imagined to do.

(a)



(b)

**FIGURE 2.** (a) The single page of chapter 1 of the Short Range Committee's report of 4 September 1959, with the verses of Genesis 11:1-6. (Source: Unprocessed Collection, NMAH, Box 97, Folder "COBOL," p. 1.) (b) Sammet's handwritten pencil notes of the table of contents of the report in question. [Source: Jean E. Sammet Papers (CBIA 77), Box 207, Folder 1, p. 229.]

elements of the biblical myth.<sup>8</sup> Not surprisingly, then, there exists a long tradition of alternative interpretations to the long-established pride and punishment reading of the biblical story and its corollary reading, the confusion of tongues. In particular, cultural historian Jonathan Sawday has shown how late 16th- and early 17th-century Flemish and Dutch painters, such as Hendrick van Cleve, Marten van Valckenborch, and Pieter Bruegel the Elder, became fascinated with the tower of Babel as a monument to human technological inventiveness.<sup>9</sup> For these artists, the biblical story was concerned not so much with linguistic confusion and dispersion as with the construction of the tower as an expression among others of technological optimism and self-confidence, human purposefulness, and virtuous industriousness. Additionally, historians of science Jim Bennet and Scott Mandelbrote have also discussed a comparable reading of the myth of Babel in mid-17th-century England, in Samuel Hartlib's intellectual circle, in which the tower appears as a symbol of "what might be preserved if people were able to cooperate with God's plans."<sup>10</sup>

Similarly, there have been voices in the history of computer programming that have also appealed to the myth of Babel to convey a message of technological optimism and celebration of the value of cooperation.

Some of these voices, as was the case of a younger Jean E. Sammet, put their hopes in the development during the late 1950s of the so-called "common" or "universal languages." These free-standing programming notations were removed from any marks that could associate them with any individual machine. This way, they were supposed to facilitate the reusability and readability of programs across different computer models, thereby reducing the costs associated with programming and making it less appealing for users and manufacturers to develop their own notations. The collaborative development of the programming languages Algol (for Algorithmic Language) and Cobol (for Common Oriented Business Language) were the first direct results of the pursuit to facilitate computer intercommunication and discourage the development of new programming notations.<sup>11</sup> And while Algol was to leave its imprint on the tower of *Communications*, the development of Cobol turned out to be the beginning of Jean E. Sammet's career-long fascination with the myth of Babel. Hence, in the first section of this article, I will focus on the tower of *Communications*, while in the second section, I will follow Sammet's gradual engagement with the two distinctive parts of Genesis 11:1-9, that is, during the development of Cobol (Genesis 11:1-6) and at the time of writing her book (Genesis 11:6-9).

<sup>8</sup>Sherman (2013); Hiebert (2007); Wolde (1994).

<sup>9</sup>Sawday, *Engines of the Imagination* (2007), pp. 20-24; Sawday, "The fortunes of Babel" (2007), pp. 202-205.

<sup>10</sup>Bennett and Mandelbrote (1998), p. 106.

<sup>11</sup>On Cobol, see: Allen (2018); Ensmenger (2010), pp. 93-101; Sammet (1981). For a brief survey of the large literature on Algol, see Nofre (2014), p. 382, footnote 11.

It is difficult to ignore the strong linguistic dimension to all these developments, right down to the name of programming *language*.<sup>12</sup> It was probably not a coincidence that Babel entered computing when programming notations were starting to be modeled on the highly abstract notion of formal language from logic. For this was a process that ran in parallel and interacted with the reconceptualization of the very idea of language in linguistics, based also upon the same notion of formal language, most notably in the work of Noam Chomsky.<sup>13</sup> But in the 1950s, concerns about language were wider than issues related solely to the techno-scientific fields of computer programming, logic, and linguistics, affecting U.S. public, political, and intellectual life.<sup>14</sup> The rise of the United States as a global power, together with the rapid growth of Soviet science (much of it in Russian), led to the belief within the political and scientific establishment of being in the midst of a “translation crisis,” in the words of historian of science Michael D. Gordin.<sup>15</sup> Computers were quickly put to work on the task of automating the translation of natural languages in order to “[erect] a new Tower of anti-Babel,” as Warren Weaver, an early advocate of machine translation, famously put it in 1955.<sup>16</sup> Machine translation projects became sites of encounter and interaction for linguists and computer programming experts, facilitating the cross-fertilization of insights, tools, and models across the fields of computer programming, linguistics, and logic.<sup>17</sup> The linguistic nature of programming notations, which was already present in the tradition of anthropomorphizing the computer, now came to be seen as even more appropriate. And the metaphor of Babel naturally joined a new constellation of metaphors (e.g. language, syntax, semantics, and translation) around which a logico-linguistic approach to the study and design of programming notations was taking shape.

### “THE DREAM OF THE PERFECT LANGUAGE”

As we look at the cover of the January 1961 issue of *Communications of the ACM*, our eyes are drawn to the tower and the gray clouds hanging above and around the tower (Figure 1). The sky is darker above the small human settlement pictured on the desert dunes at the bottom right of the tower. Construction

workers, sketched as little human figures, are busy moving around materials with the help of ropes, pulleys, a few camels, and two oxen pulling a cart. The artist has chosen to depict the moment when the tower is under construction and filled with construction workers. These two features also characterize early modern Protestant artworks of the biblical myth, functioning as a celebration of technological innovation and human industry.<sup>18</sup> The indistinctness of the construction workers contrasts with the prominence of the names of the seventy-three programming languages, tools, and systems inscribed in the bricks of the tower; equally prominent is the title of the symposium, the contributions to which formed the basis of this special issue. The programming language Algol appears at the top of the tower, visibly at quite a distance from the rest of languages inscribed on the bricks of the tower. At the base of the tower an inscription reads in capital letters, “Babel 1960,” as if it were a photograph with the place name and date written on it.

The tower on the cover of *Communications* sought to showcase the explosion in research that had transformed the field of computer programming in the years preceding the publication of the tower, roughly between 1955 and 1960. Only three years old by the time of publication of this famous cover, *Communications* had already established itself as the most prominent publication in computing and a forum for the exchange of information on technical innovations and professional developments.<sup>19</sup> The special issue in which the tower of Babel of languages appeared featured the papers contributed to the ACM Compiler Symposium held on 17 and 18 November 1960 at the National Bureau of Standards in Washington, DC, USA. This two-day event had been sponsored by the ACM Committee on Programming Languages, whose chair was Alan J. Perlis (Carnegie Institute of Technology), editor in chief of *Communications* and as such responsible for the cover of *Communications*.

This was the second symposium organized by the ACM Committee on Programming Languages dedicated to a “very special field of interest,” and both fields of interest were to feature prominently in the tower.<sup>20</sup> The first field, which concerned the development of techniques for the manipulation of nonnumerical symbols in computer programs, had been the subject of a first conference on 20 and 21 May 1960

<sup>12</sup>Nofre, Priestley, and Alberts (2014).

<sup>13</sup>Kibbee (2010); Tomalis (2006).

<sup>14</sup>Martin-Nielsen (2010).

<sup>15</sup>Gordin (2017), p. 226.

<sup>16</sup>Weaver (1955), p. vii.

<sup>17</sup>Léon (2015); Martin-Nielsen (2012); Hutchins (2000).

<sup>18</sup>Sawday, *Engines of the Imagination* (2007), p. 21; Sawday, “The fortunes of Babel” (2007), pp. 203–204.

<sup>19</sup>On the important role that the *Communications of the ACM* played in shaping the discipline of computer science, see the different contributions to Misa (2016).

<sup>20</sup>Comm. ACM (1961).





**FIGURE 3.** Jean E. Sammet pictured holding her book *Programming Languages: History and Fundamentals*. Photograph taken in Sammet's office at IBM. Exact date unknown, c. June 1969. [Source: Jean E. Sammet Papers (CBIA 77), Box 276, Folder 2.]

at the University of Pennsylvania. The second field of interest, which was the development of efficient and cost-effective compilers, was the focus of the special issue in which the tower of Babel appeared. Most of the contributions to this second symposium dealt specifically with the design of compilers for the programming language Algol 60. These contributions explored technical questions such as the handling of the bracket structure of instructions in Algol 60, the allocation of storage during the compilation of Algol 60, or the use of tree structures in the design of compilers for Algol 60.<sup>21</sup> Most of the rest of the contributions dealt either with new compiling techniques developed with Algol 60 in mind or with the challenges related to the implementation of Algol-like languages, such as the University of Michigan's MAD language and the System Development Corporation's CLIP language.<sup>22</sup>

The centrality of Algol in this special issue had its visual correspondence in the tower pictured on the cover. The tower displays clearly the name of Algol at its top and at a distance from the rest of names inscribed in the tower. The display of Algol at the top of the tower might also explain why the tower on the cover was left unfinished. In the short span of three years, the development of the first two versions of Algol, called Algol 58 and Algol 60, played a crucial role in transforming

computer programming into a proper body of knowledge.<sup>23</sup> Yet by the time of the publication of the special issue of *Communications*, the Algol project was starting to face serious difficulties. After the release in May 1960 of Algol 60, which contained many features incompatible with the first version of Algol, IBM's scientific user group SHARE threatened to withdraw its previous endorsement of Algol as standard.<sup>24</sup> At the same time, IBM was slowing down the development of Algol compilers for its computer models, aware of the threat that "common languages" like Algol and Cobol, which promised to bring about the cross-compatibility of computer programs, posed to its business model based on customer locking.

IBM and SHARE's second thoughts about Algol put pressure on the international community behind the project. In the summer of 1961, tensions had reached such a pitch that Perlis, who had played an important role in the early days of the Algol project, had come to the conclusion that the language had already served its purposes. For Perlis, Algol was now "dead," as he privately confided to SHARE Vice President Frank Engel.<sup>25</sup> But if Algol was indeed dead, something better might take place at the top of the tower in the not too distant future. This was, at least, the hope of Perlis, for when some years later, he was asked about the three dreams of computer science, Perlis' first choice was "[t]he Dream of the Perfect Language," a language that would allow for the expression of all algorithms "in a precise, elegant and transparent form."<sup>26</sup> Perlis dreamed of a language that would be capable of establishing a neat correspondence between words and the computational processes that these words were supposed to describe. In this regard, Perlis' dream resembled 17th century schemes to define an universal language that could overcome the arbitrary division between words and things, a resemblance that did not go unnoticed at that time.<sup>27</sup>

If we move down from the top of the tower, there seems at first sight to be a fair amount of arbitrariness in the arrangement of the rest of languages. Yet some

<sup>23</sup>Priestley (2010), Ch. 9.

<sup>24</sup>Nofre (2021), p. 400.

<sup>25</sup>Engel was reporting to SHARE President Harry Cantrell about his conversations with Perlis. Source: Frank Engel, Jr. to Harry Cantrell, Aug. 1, 1961, SHARE Records, Archives Center, National Museum of American History [hereafter NMAH], Box 52, Folder 4.

<sup>26</sup>A. J. Perlis to P. J. Davis, Sept. 29, 1978, Alan J. Perlis Papers (CBIA 64), Charles Babbage Institute Archives [hereafter CBIA], Box 1, Folder 50.

<sup>27</sup>Bennett and Mandelbrote (1998), p. 105. This parallelism was made explicit by Herbert Ohlman in October 1961 during a two-day meeting on the design of computer languages for information retrieval. See Ohlman (1962), p. 9.

<sup>21</sup>Grau (1961); Bottenbruch (1961); Sattley (1961); Irons and Feurzeig (1961); Huskey and Wattenburg (1961).

<sup>22</sup>Englund and Clark (1961); Arden, Galler, and Graham (1961); Irons (1961); Ingberman, "Thunks" (1961); Ingberman, "Comments" (1961).

significant choices were made that point to some kind of thought behind the ranking of some of the acronyms spread across the tower. In particular, the selection of languages at the upper part of the tower seems to hint at a celebration of progress and breakthroughs in the field and a call to expand the range of applications of computer programming. Yet before we move on to that, it is important to note that many of the “languages” ranked in the tower were in fact compilers, assemblers, early operating systems, or simply new programming techniques, which reminds us of the semantic fluidity of the notion of “language” at that time.

Indeed the first “language” that we encounter in the tower just below Algol, GP (for Generalized Programming), was defined by its developers as a “programming system.”<sup>28</sup> GP was conceived by two promising young researchers, Anatol W. Holt and William J. Turanski, who in the late 1950s were working under the direction of John Mauchly at the Univac Applications Research. Holt and Turanski’s GP was essentially a generalized filing system initially developed for the Univac I computer. GP’s library of programs consisted of pre-tested subroutines that could be automatically filed away and later called in by future programs, allowing the system to grow and become “general purpose.”<sup>29</sup> Holt and Turanski were praised for the building-block philosophy underlying GP’s expandable and flexible library of programs, which was at that time regarded “a most important development in digital computer programming.”<sup>30</sup> Holt and Turanski were also praised for being one of the first groups to give serious attention to the problems related to the structure of programs.<sup>31</sup> In 1958, Remington Rand renamed GP with the more commercial name Fleximatic. Tellingly, this name appears on the tower seven levels below GP, as if the new commercial branding of a research-oriented product did not deserve a better fate in the tower.

The high ranking conferred to GP may also have been a way of remembering Turanski’s promising but short career. Turanski had passed away one year before, at the age of 35, of injuries received after being struck by a car. The accident occurred the day before Turanski was scheduled to depart for Paris to attend the international Algol conference that would lead a few months later to the definition of Algol 60.<sup>32</sup> Turanski’s death came as a shock to the computer programming research community in the United States. Some

of its most prominent members, including Alan J. Perlis, were very active within the ACM and knew Turanski well from working with him in the process of definition of Algol and in the development of programming tools for the Fielddata Project of the U.S. Army Signal Corps.<sup>33</sup> This was not the first time that *Communications* rendered a tribute to Turanski. The May 1960 issue of *Communications* had featured a big picture of Turanski on the cover and included a one-page-long obituary placed before the first page of the soon-to-be-famous report containing the definition of Algol 60, a report which was also “[d]edicated to the Memory of William Turanski.”<sup>34</sup>

The “languages” ranked in the tower just under GP reflected how significant the pursuit of generality had become in computer programming. At this level we encounter several techniques and languages for symbolic computation, a research field whose development was necessary for a fuller realization of the potentialities of the computer in areas such as artificial intelligence and nonnumerical mathematical computation.<sup>35</sup> One of the better remembered is COMIT, a language for the automatic translation of natural languages developed by Victor Yngve’s team of linguists at the MIT Research Laboratory of Electronics.<sup>36</sup> COMIT was conceived to facilitate the search and extraction of words and structures in a text written, for example, in German and then map them into the words and structures of English, and vice versa. More generally, COMIT provided an effective means for searching and performing transformations upon a particular string of symbols. This feature explains why COMIT’s initial user base of linguists expanded very quickly to other application areas such as information retrieval and theorem proving in mathematics, areas in which the logical manipulation of strings of symbols also played a significant role.

<sup>28</sup>Holt (1958).

<sup>29</sup>Holt (1958), p. 8.

<sup>30</sup>Carr (1960).

<sup>31</sup>Rosen (1964), p. 3.

<sup>32</sup>“Obituary” (1960).

<sup>33</sup>The Fielddata program (1956–1962) sought to develop a family of compatible computers to be employed in logistics, administration, intelligence, and especially combat operations. Alan J. Perlis (Carnegie Institute of Technology), Saul Gorn (University of Pennsylvania), and Anatol Holt and William Turanski (Remington Rand), worked on the development of advance computer programming techniques, including the design of an Automatic Programming Language, which was never completed but was supposed to be universal, expandable, and capable of growth. See: Lipton (1960); Luebbert and Collom (1959).

<sup>34</sup>Backus et al. (1960).

<sup>35</sup>Stephanie Dick has noted that the development of techniques for symbolic computation played an important role in the articulation and recognition of computer science. See: Dick (2016), p. 62.

<sup>36</sup>Yngve (1961); Yngve (1960).

Another technique for symbol manipulation was also given a prominent place at the top of the tower. Placed next to COMIT, we find the acronym TRIE, which was a reference to the so-called “trie memory,” an early example of hierarchical tree data structure. While working at the U.S. Naval Ordnance Laboratory at Corona in California, Rene De La Briandais developed this nonsequential searching technique to reduce search times when using magnetic tapes, disks, and drums.<sup>37</sup> These external storage devices were slower in operation than the computer’s own working memory, yet they were commonly used in applications that required constant access to large data files, as was often the case in the areas of artificial intelligence and information retrieval. Edward Fredkin developed much the same technique at about the same time and independently and gave it the name of “trie,” a play on the words “retrieval” and “tree.”<sup>38</sup>

De La Briandais was inspired by the work of Allen Newell and John Clifford Shaw at the RAND Corporation, whose IPL language appeared ranked in the tower below the trie memory, as was the case too with John McCarthy’s LISP language.<sup>39</sup> This ranking may be due to the fact that for many search operations, IPL and LISP required often iterating through all the elements in a list of symbols, making them slower for many applications. Other similar sequences in increasing technical sophistication also found a visual correspondence in the tower. This is, for example, the case for the Fortran family of languages, with new and more sophisticated versions, or subsets, of the original Fortran language, such as XTRAN, Fortran II, Fortran III, and Fortransit, ranked in upward order above the original Fortran language. But perhaps what best reflects the sense of progress and advance that the tower sought to convey is that at the bottom of the tower we find some of the earliest forms of cooperation among computing installations in the development of common programming tools (i.e., PACT-I, SAP, and USE).

## GENESIS 11:1-9

Jean E. Sammet’s fascination with the myth of Babel started early on in her career when in the spring of 1959 she joined the U.S. Department of Defense’s initiative to develop a “common business language,” an initiative that would lead one year later to the first

definition of Cobol.<sup>40</sup> The development of Cobol is a familiar narrative in the history of computing, not in small part due to Sammet’s own later historiographical work.<sup>41</sup> But our reliance on Sammet’s own work as a historian has somehow concealed her early views on the multiplicity of programming languages. Back in 1959, Sammet’s approach to the question of the proliferation of languages was much more confident and embraced a much more optimistic interpretation of the myth of Babel than the vision that would infuse her seminal monograph *Programming Languages: History and Fundamentals* ten years later.

Originally trained as a mathematician at the University of Illinois, Sammet’s career in computing started in 1955 at Sperry Gyroscope.<sup>42</sup> At the time she joined the Cobol initiative, Sammet led at Sylvania Electric Products the development of programming tools for Sylvania’s MOBIDIC computer, as part of the U.S. Army’s Fielddata project.<sup>43</sup> The first official steps of the Cobol project started on 18 May 1959, when Charles A. Phillips, then director of the data systems research staff from the Office of the Assistant Secretary of Defense, convened a now famous meeting at the Pentagon on 28 and 29 May.<sup>44</sup> This meeting involved “a restricted nucleus of well-qualified key people,” representing both users and manufacturers, “to consider the development of specifications for a common business language.”<sup>45</sup> As head of the office concerned with the use of computers for data processing purposes in the three military departments, Phillips had very specific interests in supporting the standardization and cross-compatibility of programming notations. Sammet’s handwritten pencil notes from the Pentagon meeting confirm the “pressure on Phillips to designate

<sup>40</sup>Conference on Data Systems Languages (1960).

<sup>41</sup>Sammet (1985); Sammet (1981). On Sammet’s crucial role in organizing the first conference on the history of programming languages in 1978, see Mittlmeier (2024).

<sup>42</sup>Bergin (2009), pp. 77-78.

<sup>43</sup>On the Fielddata project, see footnote 33.

<sup>44</sup>One month earlier Mary K. Hawes (Burroughs) had convened a meeting at the University of Pennsylvania to discuss plans “to develop the specifications for a common business language”. Following a suggestion from Grace M. Hopper (Remington Rand), the participants at this meeting agreed that the Department of Defense through Charles Phillips should be asked to sponsor the first formal meeting to “carry more weight and [...] create sufficient pressure to enable rapid development of the entire project”. Source: I. E. Block, “Conference report. Preliminary meeting for the organization of a common business language”, University of Pennsylvania Computing Centre, Apr. 8, 1959, Computer Standards Collection, 1958-1978, NMAH, Box 10, Folder ‘COBOL-History’, Volume 3, p. 2, 4.

<sup>45</sup>C. A. Phillips to Frank Middleswart (I and E. Dupont), May 18, 1959, Unprocessed Collection, NMAH, Box 97, Folder ‘COBOL’.

<sup>37</sup>De La Briandais (1959).

<sup>38</sup>Fredkin (1960).

<sup>39</sup>De La Briandais (1959), p. 297. On the work of Newell and Shaw at RAND, see Dick (2015), pp. 629-634.

one of 3 mil. [military] agencies to standardize ADP [automatic data processing]."<sup>46</sup>

The standardization of "billing and computing machines" within the U.S. Navy and the U.S. Air Force had already been scheduled to take place in the fiscal years of 1957 and 1958, though the development of detailed plans had not yet started by June 1956. These efforts were part of the Defense Standardization Program started by the Department of Defense at the behest of Congress in 1952.<sup>47</sup> Logistics drove this major effort in military standardization, which had as one important goal the reduction of the variety of military equipment items by adopting the "common language" of the federal cataloguing program.<sup>48</sup> Pursuing efficiency through simplicity was also the Department of Defense's approach to the Cobol initiative. As Phillips explained in detail to the participants at the Pentagon meeting, the most pressing outcome of the rapid growth in the number of computers was the equally rapid growth in personnel costs.<sup>49</sup> These costs were particularly high in the early period after the delivery of a computer, when it was estimated that two-thirds of personnel costs went to programming staff, while the costs on a continuing basis were about one-third.<sup>50</sup> Hence, the Pentagon meeting had as a goal to explore the possibility of establishing a "standard POL [Problem Oriented Language]," which would allow computing facilities to save time and personnel costs invested in rewriting programs to go from one kind of equipment to another.<sup>51</sup>

<sup>46</sup>Sammet, "Miscellaneous Papers re May 28-29, 1959 meeting in Pentagon including my handnotes", May 28-29, 1959, Jean E. Sammet Papers (CBIA 77), Box 207, Folder 1, p. 2.

<sup>47</sup>The Defense Standardization Program was part of the "Defense Cataloguing and Standardization Act", 82nd Congress, Public Law 436, July 1952 (later codified as Public Law 1028, 84th Congress, Chapter 145). On plans from the US Navy and the US Air Force to standardize billing and computing machines, see: U.S. House, Committee on Armed Services (1956), p. 34, footnote 1.

<sup>48</sup>McDonald (2010), p. 33; US House, Committee on Appropriations (1956), p. 172.

<sup>49</sup>The three military departments together possessed by September 1958 a total of 116 installed computers, in addition to 187 on order or planned. Phillips expected that the total number of installed computers would have reached the figure of 300 by the end of 1959. These figures excluded the computers in weapon systems, weapon testing systems, the SAGE system, and those from the civilian agencies of the US Department of Defense. Source: "Miscellaneous Papers re May 28-29, 1959 meeting in Pentagon including my own handnotes", May 28-29, 1959, Jean E. Sammet Papers (CBIA 77), Box 207, Folder 1, p. 1.

<sup>50</sup>"Miscellaneous Papers re May 28-29, 1959 meeting in Pentagon including my own handnotes", May 28-29, 1959, Jean E. Sammet Papers (CBIA 77), Box 207, Folder 1, p. 1.

<sup>51</sup>"Miscellaneous Papers re May 28-29, 1959 meeting in Pentagon including my own handnotes", May 28-29, 1959, Jean E. Sammet Papers (CBIA 77), Box 207, Folder 1, p. 2.

The participants of the Pentagon meeting agreed to establish a permanent organization, and by the end of June the Committee (also variously referred to as "Conference") on Data Systems Languages (CODASYL) had been established.<sup>52</sup> Three task groups were also established, called the Short Range, Intermediate Range, and Long Range Committees, each one focusing on a different time scale and approach to the goal of establishing a "common business language." Sammet became the third woman of the initial Short Range Committee, the other two women being Mary K. Hawes (Burroughs) and Frances E. Holberton (USN David Taylor Model Basin).<sup>53</sup> Under the leadership of Joseph H. Wegstein (National Bureau of Standards), the Short Range Committee was to assess "present languages and their faults" and "come up with a recommendation as to a common language" within three months.<sup>54</sup>

Yet what was supposed to be an assessment exercise of the various qualities of existing languages turned very quickly into the development of an entirely new language. Sammet played an important role in facilitating this shift in policy during the many meetings of the Short Range Committee that took place over the months of July and August. The main thrust of the committee's work in June and early July went to study the data description and procedural statements of three languages that were then under development, namely, IBM's Comtran (Commercial Translator), Univac's Flowmatic, and USAF Air Materiel Command's Aimaco (Air Material Compiler). However, by the end of July, the Short Range Committee had already shifted its priorities to developing a new language. During this period of intense work, Sammet and Hawes took a leading role in the committee's deliberations, as well as in the process of writing down the specifications of the new language-in-the-making. Hawes, Sammet, and Wegstein did also manage to overcome the resistance of IBM's representatives to the committee and produce a report by the end of August presenting a "framework upon which an effective Common Business Language [could] be built."<sup>55</sup>

<sup>52</sup>Sammet (1981), p. 203.

<sup>53</sup>The complete list of representatives to the Short Range committee at one time or another is to be found in: "Cobol, 7 Nov. 1959", Jean E. Sammet Papers (CBIA 77), Box 207, Folder 20, p. c. On the question of women's representation in the early development of Cobol, see: Abbate (2012), pp. 80-81.

<sup>54</sup>J. Sammet to N. Zachery, Memo: Pentagon Conference on Common Business Language, 3 June 1959, in "Miscellaneous Papers re May 28-29, 1959 meeting in Pentagon including my own handnotes", May 28-29, 1959, Jean E. Sammet Papers (CBIA 77), Box 207, Folder 1, p. 3.

<sup>55</sup>"A Report to the Executive Committee of the Committee on Data Systems Languages by the Fact-Finding and Short Range Language Committee, September 4, 1959", Unprocessed Collection, NMAH, Box 97, Folder 'COBOL', p. 2.



On 9 September the Executive Committee approved the report of the Short Range Committee, which tellingly now styled itself as the “PDQ committee.” The report already included a description of the character set and data of the future “Common Business Language,” along with the description of the syntax of its “verbs” and a first attempt to define their semantics. The report also acknowledged the rough character of the language description. Hence, the Short Range Committee requested authorization from the Executive Committee “to complete and polish the system by December 1, 1959.”<sup>56</sup> But perhaps what best encapsulated the bold agenda of the Short Range Committee was the first chapter of the report, which had as its title “GENESIS,” in uppercase, and consisted only of the quotation of verses Genesis 11:1-6, with the words “one language” underlined [Figure 2(a)]<sup>57</sup>:

1. And the whole earth was of “one language” [sic], and of one speech.
2. And it came to pass, as they journeyed from the east, that they found a plain in the land of Shinar; and they dwelt there.
3. And they said one to another, Go to, let us make brick, and burn them thoroughly. And they had brick for stone, and slime had they for mortar.
4. And they said, Go to, let us build us a city, and a tower, whose top may reach unto heaven; and let us make a name, lest we be scattered abroad upon the face of the whole earth.
5. And the LORD came down to see the city and the tower, which the children of men builded.
6. And the LORD said, Behold the people is one, and they have all one language; and this they begin to do; and now nothing will be restrained from them, which they have imagined to do.

The decision to include verses Genesis 11:1-6 in the report seems to have been made during the second day of a two-day meeting of the Short Range Committee on 27 and 28 August 1959. Sammet’s notes of this meeting contain the agreed table of contents of the report, including the title of the report’s first chapter, “Genesis,” followed by the words “11: 1-6, King James version”

[Figure 2(b)].<sup>58</sup> Such a choice was a sign of the self-confidence and technological optimism with which the members of the committee, including Sammet, contemplated their task of bringing about an era of “one language.”

Ten years later, optimism had given way to a more pragmatic outlook. Sammet had joined IBM in 1961 with the task of creating a new research group focused on advanced development work in programming, which became known as the Boston Advanced Programming Department, part of the IBM Systems Development Division.<sup>59</sup> In June 1966, Sammet began work on the main portion of *Programming Languages: History and Fundamentals*, following an invitation from George Forsythe, head of the department of computer science at Stanford and editor of the Prentice-Hall Automatic Computation Series, to write a book about programming languages.<sup>60</sup> By this time, Sammet had come to regard the large quantity of programming languages as something inevitable and something that should not be fought against by means of uniformization. As she put it in her book, the development of a truly universal programming language was not only “unachievable in the foreseeable future” but also not “desirable.”<sup>61</sup> Such a prospect would only prevent progress and “force regimentation of an undesirable and impractical kind” and inevitably lead to “deviations” from the universal language, which would take us back to square one.<sup>62</sup>

As the book offer from Prentice-Hall put it, *Programming Languages: History and Fundamentals* sought to alleviate the “complexity,” “confusion,” and “miscomprehension” that had arisen in the field of programming.<sup>63</sup> In keeping with this purpose the book provided in one volume and in a consistent fashion an extensive amount of technical and historical information on all the major and most of the minor programming languages developed in the United States up to the fall of 1967, information that Sammet had painstakingly

<sup>56</sup>“A Report to the Executive Committee of the Committee on Data Systems Languages by the Fact-Finding and Short Range Language Committee, September 4, 1959”, Unprocessed Collection, NMAH, Box 97, Folder ‘COBOL’, p. 2.

<sup>57</sup>“A Report to the Executive Committee of the Committee on Data Systems Languages by the Fact-Finding and Short Range Language Committee, September 4, 1959”, Unprocessed Collection, NMAH, Box 97, Folder ‘COBOL’, p. 1.

<sup>58</sup>“Miscellaneous Papers re May 28-29, 1959 meeting in Pentagon including my own handnotes”, May 28-29, 1959, Jean E. Sammet Papers (CBIA 77), Box 207, Folder 1, p. 229.

<sup>59</sup>Bergin (2009), p. 79.

<sup>60</sup>IBM, “[Untitled: Draft of book’s press release]”, n.d., Jean E. Sammet Papers (CBIA 77), Box 276, Folder 2, p.1. See also: J. Abbate, “Oral-History: Jean Sammet.” ETHW.org. Accessed: Mar. 1, 2024. [Online] Available: [https://ethw.org/Oral-History:Jean\\_Sammet](https://ethw.org/Oral-History:Jean_Sammet)

<sup>61</sup>Sammet (1969), p. 723.

<sup>62</sup>Sammet (1969), p. 723.

<sup>63</sup>Prentice Hall, “Programming Languages: History and Fundamentals, book offer – Prentice Hall, Inc. 1970”, Jean E. Sammet Papers (CBIA 77), Box 11, Folder 26, p. 4.

collected over the years.<sup>64</sup> The detailed table of contents of the book was aimed to show “the structure of the programming language field at glance.”<sup>65</sup> The large amount of reference material that the book contained (over 800 items) was consistent with Sammet’s purpose of producing an “encyclopedic” book to “serve as a starting reference for all languages.”<sup>66</sup> So the reader who consulted it to learn more about an existing language would not have to search different sources throughout which the information was scattered. A detailed description of the historical development of the major languages also provided the reader perspective and a better understanding to “the growing Tower of Babel in the programming world.”<sup>67</sup> Moreover, Sammet’s new and more resigned views about the proliferation of languages were already announced in the book’s epigraph, which quotes Genesis 11:6-9, the verses that have traditionally provided the basis for the pride-and-punishment interpretation of the biblical story:

6. And the LORD said, Behold, the people is one, and they have all one language; and this they begin to do; and now nothing will be restrained from them, which they have imagined to do.
7. Go to, let us go down, and there confound their language, that they may not understand one another’s speech.
8. So the LORD scattered them abroad from thence upon the face of all the earth; and they left off to build the city.
9. Therefore is the name of it called Babel; because the LORD did there confound the language of all the earth; and from thence did the LORD scatter them abroad upon the face of all the earth.

Sammet’s new pragmatism also found expression in the tower of Babel that features in the front endpaper and the book jacket (Figure 3). Set against a medium to light blue background, the tower of Babel on the cover of Sammet’s book, which names all the languages described in the book, seems to have been sketched rather quickly, as if the artist were merely attempting

to give the outlines of the tower.<sup>68</sup> The book acknowledges the illustrator of popular science and children’s books George Bakacs as the author of the “jacket design.”<sup>69</sup> But the clean lines and colorful palettes that characterized Bakacs’ work as an illustrator differed considerably from the drawing style and the white and black colors of the tower. This suggests the possibility that Sammet herself might have contributed in one way or another to the design of the cover. In fact, in 2011, Sammet told historian of computing Thomas J. Bergin that she had brought the January 1961 issue of *Communications* and a list of the languages in the book to an artist at IBM, with whom she discussed the idea of a tower of Babel.<sup>70</sup>

In any case, Sammet’s tower comes across as a vulnerable construction, with little left on it of the confident ambition and celebration of progress that had characterized the tower of *Communications*, which had been her inspiration. Also unlike *Communications*’ tower, there is no other hierarchy in the arrangement of the languages in Sammet’s tower than the encyclopedic alphabetical order, an expression of the author’s determination to bring order into a confused situation.<sup>71</sup> Meanwhile, the workers that fill the ramps of the tower keep carrying new bricks to continue building a tower that seems to have no end, just as there seems to be no end to the “plethora of languages” being developed.<sup>72</sup> And so as not to leave any doubt about the complexity into which the field of programming languages had grown, Sammet’s tower is crowned with the name “Babel” in bold capital letters.

## AFTERLIFE

In the 1960s it became customary to talk of the “burgeoning ‘tower-of-Babel’ phenomenon” in the field of programming languages.<sup>73</sup> Yet contrary to the iconic status that the tower of *Communications* has enjoyed in recent decades, this image aroused little comment at the time of its publication. In the few ex-

<sup>64</sup>The encyclopedic character of the book is one aspect that most of the reviewers appreciated. See, for example: Rosen (1971); Abrahams (1970).

<sup>65</sup>Prentice Hall, “Programming Languages: History and Fundamentals, book offer – Prentice Hall, Inc. 1970”, Jean E. Sammet Papers (CBIA 77), Box 11, Folder 26, p. 2.

<sup>66</sup>IBM, “[Untitled: Draft of book’s press release]”, n.d., Jean E. Sammet Papers (CBIA 77), Box 276, Folder 2, pp. 1-2.

<sup>67</sup>IBM, “[Untitled: Draft of book’s press release]”, n.d., Jean E. Sammet Papers (CBIA 77), Box 276, Folder 2, p.1.

<sup>68</sup>There are in total one hundred seventeen names in Sammet’s tower, provided that we count A-2 and A-3 as one language, as Sammet does in her book.

<sup>69</sup>See, for example: Schere and Bakacs (1969); Silverstein, Silverstein, and Bakacs (1970); Silverstein, Silverstein, and Bakacs (1969).

<sup>70</sup>Author’s personal communication with Thomas J. Bergin, Dec. 18, 2011.

<sup>71</sup>All the languages are arranged in descending alphabetical order according to their first letter with the exception of the language Klerer-May, which is ranked between letters G and I.

<sup>72</sup>Prentice Hall, “Programming Languages: History and Fundamentals, book offer – Prentice Hall, Inc. 1970”, Jean E. Sammet Papers (CBIA 77), Box 11, Folder 26, p. 2.

<sup>73</sup>Ledley and Wilson (1962), p. 145.

isting references, the tower is characterized as being “joyfully depicted” or “amusing,” but the image was not really seen.<sup>74</sup> Rather, the traditional reading of the biblical story was projected onto the tower of *Communications* by means of a comparison between the large quantity of programming languages and “the senseless repetition of human language chaos” that was now “perpetrated upon the new machines.”<sup>75</sup> Such a “confusion of tongues” interpretation of the tower on the cover of *Communications* would become commonplace, particularly after the publication of Sammet’s book.

The tower of Babel that featured on the cover of Sammet’s *Programming Languages* drew again attention to the tower of *Communications*. But at the same time, Sammet’s tower eclipsed the message of technological optimism and self-confidence in the development of computer programming that the journal’s cover had sought to convey. Since then, these two images have become conflated and transformed into a symbol of “our sense of frustration” and “confusion at the proliferation of programming languages.”<sup>76</sup> Sammet never made any public remark that could be interpreted as her dissatisfaction with the way her tower had been received. Quite the opposite, Sammet took the opportunities presented to her to proudly celebrate her book’s cover, which was visibly on display at the landmark exhibition “Information Age: People, Information and Technology” held at the Smithsonian’s National Museum of American History between 1990 and 2006.

Looking back in 1991 on a long-career engagement with the history of programming languages, Sammet published a second tower of Babel, which she had drawn herself representing the state of the field in the early 1970s.<sup>77</sup> Loaded with more than 160 languages, this second tower is about to fall, were it not for the fact that it rests on a ladder inscribed with the names of the most enduring languages (i.e., Fortran and Cobol). Probably, Sammet intended to use this second tower as the cover for the second, updated edition of her book, which she worked on for several years but never completed.

In this article I have contextualized the iconic towers on the covers of *Communications* and Sammet’s book and their stories on their own terms. These images were a response to the specific concerns, aspirations, and expectations of their creators. They should

thus not be read as sources that simply proclaim the abundance of programming notations. Instead, I have reconstructed what these images tell us, even if at first they seem to challenge well-established views on the question of the proliferation of programming notations. But the prominence of the myth of Babel in computer programming also testifies to the powerful centrality of the language metaphor in articulating the field during the 1960s. Further research may elucidate to what extent such a centrality was related to our eagerness to find a way to talk to computers. Yet this also means that notions such as that of “one language,” or even the “perfect language,” deserve more attention in the historiography, particularly as catalysts of research during this period, rather than being dismissed, ignored, or condemned as utopian dreams.

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## BIBLIOGRAPHY

- [1] J. Abbate, *Recoding Gender: Women’s Changing Participation in Computing*. Cambridge, MA, USA: MIT Press, 2012.
- [2] P. Abrahams, “[Review of programming languages: History and fundamentals, by J. E. Sammet],” *Math. Comput.*, vol. 24, no. 110, pp. 500–501, Apr. 1970, doi: [10.2307/2004521](https://doi.org/10.2307/2004521).
- [3] B. Allen, “Common language: Business programming languages and the legibility of programming,” *IEEE Ann. Hist. Comput.*, vol. 40, no. 2 pp. 17–31, Apr./Jun. 2018, doi: [10.1109/MAHC.2018.022921441](https://doi.org/10.1109/MAHC.2018.022921441).
- [4] Archives Center, National Museum of American History, Smithsonian Institution, Washington, DC, USA.
- [5] B. W. Arden, B. A. Galler, and R. M. Graham, “The internal organization of the MAD translator,” *Commun. ACM*, vol. 4, no. 1, pp. 28–31, Jan. 1961, doi: [10.1145/366062.366079](https://doi.org/10.1145/366062.366079).
- [6] T. K. Astarte, “Formalizing meaning: A history of programming language semantics,” Ph.D. dissertation, School of Comput., Newcastle Univ., Newcastle upon Tyne, U.K., 2019.

<sup>74</sup>Goldstein (1965), p. 9; Ohlman (1962), p. 9.

<sup>75</sup>Ohlman (1962), p. 9.

<sup>76</sup>Horowitz (1984), p. xii; Wegner (1970), p. 13.

<sup>77</sup>Sammet (1991), p. 42.

- [7] J. W. Backus et al., "Report on the algorithmic language ALGOL 60," *Commun. ACM*, vol. 3, no. 5, pp. 299–311, May 1960, doi: [10.1145/367236.367262](https://doi.org/10.1145/367236.367262).
- [8] J. A. Bennett and S. Mandelbrote, *The Garden, the Ark, the Tower, the Temple: Biblical Metaphors of Knowledge in Early Modern Europe*. Oxford, U.K.: Museum of the History of Science, Bodleian Library, 1998.
- [9] T. J. Bergin, "Jean Sammet: Programming language contributor and historian, and ACM President," *IEEE Ann. Hist. Comput.*, vol. 31, no. 1, pp. 76–85, Jan./Mar. 2009, doi: [10.1109/MAHC.2009.14](https://doi.org/10.1109/MAHC.2009.14).
- [10] Bodleian Libraries, Univ. of Oxford, U.K.
- [11] H. Bottenbruch, "Use of magnetic tape for data storage in the ORACLE-ALGOL translator," *Commun. ACM*, vol. 4, no. 1, pp. 15–19, Jan. 1961, doi: [10.1145/366062.366075](https://doi.org/10.1145/366062.366075).
- [12] J. W. Carr III, "[Review] Bruno Thüring, Einführung in die Methoden der Programmierung kaufmännischer und wissenschaftlicher Probleme für elektronische Rechenanlagen," *Commun. ACM*, vol. 3, no. 2, Feb. 1960, Art. no. 120.
- [13] Charles Babbage Institute Archives, Univ. of Minnesota, Minneapolis, MN, USA.
- [14] *Commun. ACM*, vol. 4, no. 1, Jan. 1961.
- [15] *Computers and Automation*, vol. 17, no. 8, Newtonville, MA, USA: Berkeley Enterprises Inc., Aug. 1968, pp. 1–27.
- [16] Conference on Data Systems Languages, COBOL: Initial Specifications for a Common Business Oriented Language. US Dept. of Defense, U.S. Govt. Printing Office, Washington, DC, USA, Apr. 1960.
- [17] R. De La Briandais, "File searching using variable length keys," in *Proc. IRE-AIEE-ACM '59 (Western): Papers Presented at the March 3–5, 1959, Western Joint Comput. Conf.*, New York, NY, USA: ACM, 1959, pp. 295–298, doi: [10.1145/1457838.1457895](https://doi.org/10.1145/1457838.1457895).
- [18] S. Dick, "Computer science," in *A Companion to the History of American Science*, G. M. Montgomery, M. A. Largent, Eds., Hoboken, NJ, USA: Wiley, 2015, pp. 55–68.
- [19] S. Dick, "Of models and machines: Implementing bounded rationality," *Isis*, vol. 106, no. 3, pp. 623–634, Sep. 2015, doi: [10.1086/683527](https://doi.org/10.1086/683527).
- [20] D. Englund and E. Clark, "The CLIP translator," *Commun. ACM*, vol. 4, no. 1, pp. 19–22, Jan. 1961, doi: [10.1145/366062.366076](https://doi.org/10.1145/366062.366076).
- [21] N. Ensmenger, *The Computer Boys Take Over: Computers, Programmers and the Politics of Technical Expertise*. Cambridge, MA, USA: MIT Press, 2010.
- [22] E. Frenkin, "Trie memory," *Commun. ACM*, vol. 3, no. 9, pp. 490–499, Sep. 1960.
- [23] M. Goldstein, "Computer languages," *Amer. Math. Monthly*, vol. 72, no. 2, pp. 141–146, 1965, doi: [10.2307/2313321](https://doi.org/10.2307/2313321).
- [24] M. D. Gordin, *Scientific Babel: The Language of Science from the Fall of Latin to the Rise of English*. London, U.K.: Profile Books, 2017.
- [25] A. A. Grau, "Recursive processes and ALGOL translation," *Commun. ACM*, vol. 4, no. 1 pp. 10–15, Jan. 1961, doi: [10.1145/366062.366074](https://doi.org/10.1145/366062.366074).
- [26] M. J. Halvorson, *Code Nation: Personal Computing and the Learn to Program Movement in America*. New York, NY, USA: ACM, 2020.
- [27] T. Hiebert, "The Tower of Babel and the origin of the world's cultures," *J. Biblical Literature*, vol. 126, no. 1, pp. 29–58, Apr. 2007, doi: [10.2307/27638419](https://doi.org/10.2307/27638419).
- [28] A. W. Holt, "General purpose programming systems," *Commun. ACM*, vol. 1, no. 5, pp. 7–9, May 1958, doi: [10.1145/368819.368851](https://doi.org/10.1145/368819.368851).
- [29] E. Horowitz, *Fundamentals of Programming Languages*. Rockville, MD, USA: Computer Science Press, 1984.
- [30] H. D. Huskey and W. H. Wattenburg, "Compiling techniques for Boolean expressions and conditional statements in ALGOL 60," *Commun. ACM*, vol. 4, no. 1, pp. 70–75, Jan. 1961, doi: [10.1145/366062.366098](https://doi.org/10.1145/366062.366098).
- [31] W. J. Hutchins, Ed., *Early Years in Machine Translation: Memoirs and Biographies of Pioneers*. Amsterdam, The Netherlands: Benjamins, 2000.
- [32] P. Z. Ingerman, "Thanks: A way of compiling procedure statements with some comments on procedure declarations," *Commun. ACM*, vol. 4, no. 1, pp. 55–58, Jan. 1961, doi: [10.1145/366062.366084](https://doi.org/10.1145/366062.366084).
- [33] P. Z. Ingerman, "Dynamic declarations," *Commun. ACM*, vol. 4, no. 1, pp. 59–60, Jan. 1961, doi: [10.1145/366062.366087](https://doi.org/10.1145/366062.366087).
- [34] E. T. Irons and W. Feurzeig, "Comments on the implementation of recursive procedures and blocks in ALGOL 60," *Commun. ACM*, vol. 4, no. 1, pp. 65–69, Jan. 1961, doi: [10.1145/366062.366090](https://doi.org/10.1145/366062.366090).
- [35] E. T. Irons, "A syntax directed compiler for ALGOL 60," *Commun. ACM*, vol. 4, no. 1, pp. 51–55, Jan. 1961, doi: [10.1145/366062.366083](https://doi.org/10.1145/366062.366083).
- [36] D. A. Kibbee (ed.), *Chomskyan (R)evolutions*. Amsterdam, The Netherlands: Benjamins, 2010.
- [37] R. S. Ledley and J. B. Wilson, "Automatic-programming-language translation through syntactical analysis," *Commun. ACM*, vol. 5, no. 3, pp. 145–155, Mar. 1962, doi: [10.1145/366862.366872](https://doi.org/10.1145/366862.366872).
- [38] J. Léon, *Histoire de L'Automatisation des Sciences du Langage*. Lyon, France: ENS éditions, 2015.
- [39] M. A. Lipton, "Automatic data processing equipment program for the Field Army," *IRE Trans. Military*

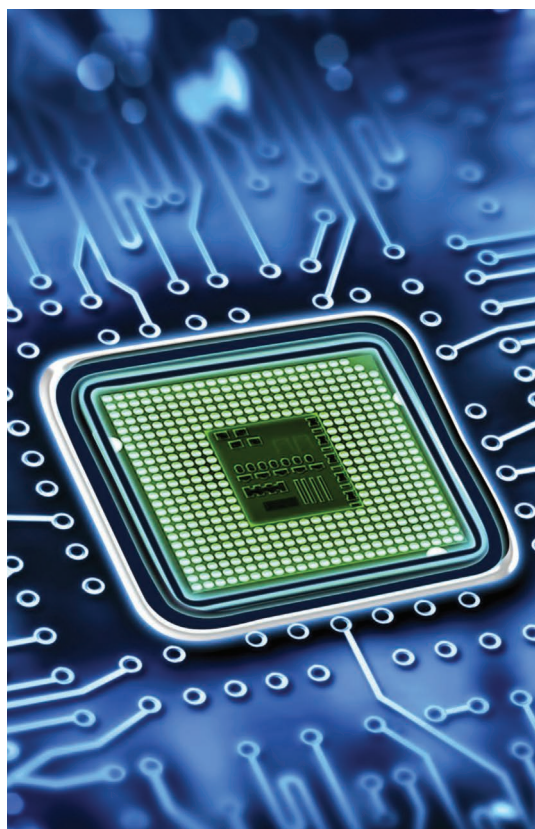


- Electron.*, vol. MIL-4, no. 4, pp. 514–518, Oct. 1960, doi: [10.1109/IRET-MIL.1960.5008281](https://doi.org/10.1109/IRET-MIL.1960.5008281).
- [40] W. F. Luebbert and P. W. Collom Jr., "Signal Corps research and development on automatic programming of digital computers," *Commun. ACM*, vol. 2, no. 2, pp. 22–27, Feb. 1959, doi: [10.1145/368280.368293](https://doi.org/10.1145/368280.368293).
- [41] J. Martin-Nielsen, "It was all connected: Computers and linguistics in early cold war America," in *Cold War Social Science: Knowledge Production, Liberal Democracy, and Human Nature*, M. Solovey, H. Cravens, Eds., New York, NY, USA: Palgrave Macmillan, 2012, pp. 63–78.
- [42] J. Martin-Nielsen, "This war for men's minds: The birth of a human science in Cold War America," *Hist. Human Sci.*, vol. 23, no. 5, pp. 131–155, Dec. 2010, doi: [10.1177/0952695110378952](https://doi.org/10.1177/0952695110378952).
- [43] C. McDonald, "From art form to engineering discipline: A history of US military software development standards, 1974–1998," *IEEE Ann. Hist. Comput.*, vol. 32, no. 4, pp. 32–47, Oct./Dec. 2010, doi: [10.1109/MAHC.2009.58](https://doi.org/10.1109/MAHC.2009.58).
- [44] T. Misa, Ed., *Communities of Computing: Computer Science and Society in the ACM*. San Rafael, CA, USA: Morgan & Claypool, 2016.
- [45] A. Mittlmeier, "History is touchy" Die history-of-programming-languages-konferenz, 1978," *Berichte Zur Wissenschaftsgeschichte*, early access, Jul. 24, 2024, doi: [10.1002/bewi.202300032](https://doi.org/10.1002/bewi.202300032).
- [46] D. Nofre, "The politics of early programming languages: IBM and the Algol project," *Hist. Studies Natural Sci.*, vol. 51, no. 3, pp. 379–413, 2021, doi: [10.1525/hsns.2021.51.3.379](https://doi.org/10.1525/hsns.2021.51.3.379).
- [47] D. Nofre, M. Priestley, and G. Alberts, "When technology became language: The origins of the linguistic conception of computer programming, 1950–1960," *Technol. Culture*, vol. 55, no. 1, pp. 40–75, Jan. 2014, doi: [10.1353/tech.2014.0031](https://doi.org/10.1353/tech.2014.0031).
- [48] "Obituary: William J. Turanski: 1925–1960," *Commun. ACM*, vol. 3, no. 5, May 1960, Art. no. A14.
- [49] H. Ohlman, "Discussion—The pros and cons of a special IR language: Pro a special IR language," *Commun. ACM*, vol. 5, no. 1, pp. 8–10, Jan. 1962, doi: [10.1145/366243.366293](https://doi.org/10.1145/366243.366293).
- [50] J. Pflüger, "Language in computing," in *Experimenting in Tongues: Studies in Science and Language*, M. Dörries, Ed., Stanford, CA, USA: Stanford Univ. Press, 2002, pp. 125–164.
- [51] M. Priestley, *A Science of Operations: Machines, Logic and the Invention of Programming*. New York, NY, USA: Springer, 2010.
- [52] S. Rosen, "Programming languages: History and fundamentals (Jean E. Sammet)," *SIAM Review*, vol. 13, no. 1, pp. 108–109, 1971, doi: [10.1137/1013011](https://doi.org/10.1137/1013011).
- [53] S. Rosen, "Programming systems and languages: A historical survey," in *Proc. Joint Comput. Conf. (AFIPS)*, New York, NY, USA: ACM, 1964, pp. 1–15.
- [54] J. E. Sammet, "Some approaches to, and illustrations of programming language history," *Ann. Hist. Comput.*, vol. 13, no. 1, pp. 33–50, Jan./Mar. 1991, doi: [10.1109/MAHC.1991.10001](https://doi.org/10.1109/MAHC.1991.10001).
- [55] J. E. Sammet, "Brief summary of the early history of Cobol," *Ann. Hist. Comput.*, vol. 7, no. 4, pp. 288–303, Oct./Dec. 1985, doi: [10.1109/MAHC.1985.10044](https://doi.org/10.1109/MAHC.1985.10044).
- [56] J. E. Sammet, "The early history of COBOL," in *History of Programming Languages*, Richard L. Wexelblat, Ed., New York, NY, USA: ACM, 1981, pp. 199–243.
- [57] J. E. Sammet, *Programming Languages: History and Fundamentals*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1969.
- [58] K. Sattley, "Allocation of storage for arrays in ALGOL 60," *Commun. ACM*, vol. 4, no. 1, pp. 60–65, Jan. 1961, doi: [10.1145/366062.366088](https://doi.org/10.1145/366062.366088).
- [59] J. Sawday, *Engines of the Imagination: Renaissance Culture and the Rise of the Machine*. London, U.K.: Routledge, 2007.
- [60] J. Sawday, "The fortunes of Babel: technology, history, and Genesis 11: 1–9," in *The Word and the World: Biblical Exegesis and Early Modern Science*, K. Killeen and P. Forshaw, Eds., Basingstoke, U.K.: Palgrave Macmillan, 2007, pp. 191–214.
- [61] M. Schere and G. Bakacs (Illustrator), *The Story of Maps*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1969.
- [62] P. M. Sherman, *Babel's Tower Translated: Genesis 11 and Ancient Jewish Interpretation*. Leiden, The Netherlands, Brill, 2013.
- [63] A. Silverstein, V. B. Silverstein, and G. Bakacs (Illustrator), *Circulatory Systems: The Rivers Within*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1970.
- [64] A. Silverstein, V. B. Silverstein, and G. Bakacs (Illustrator), *The Respiratory System: How Living Creatures Breathe*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1969.
- [65] M. Tomalin, *Linguistics and the Formal Sciences: The Origins of Generative Grammar*. Cambridge, U.K.: Cambridge Univ. Press, 2006.
- [66] U.S. House, 84th Congress, 2d Session, Committee on Armed Services, Subcommittee for Special Investigations, *Eight Report to Congress (January 1, 1956, Through June 30, 1956) of the Department of Defense: In Accordance with Section 8 of the Defense Cataloguing and Standardization Act of 1952, Public Law 436, 82d Congress*, U.S. Govt. Printing Office, Washington, DC, USA, 1956.
- [67] U.S. House, 84th Congress, 2d Session, Committee on Appropriations, *Department of Defense*

*Appropriations for 1957: Office of the Secretary of Defense, Interservice Activities, Language Changes: Hearings Before the Subcommittee of the Committee on Appropriations*, U.S. Govt. Printing Office, Washington, DC, USA, 1956.

- [68] W. Weaver, "Foreword. The new tower," in *Machine Translation of Languages: Fourteen Essays*, W. N. Locke and D. A. Booth, Eds., Cambridge, MA, USA: MIT Press, 1955, pp. v–vii.
- [69] P. Wegner, "Three computer cultures: Computer technology, computer mathematics, and computer science," in *Advances in Computers*, F. L. Alt and M. Rubinoff, Eds., vol. 10, Amsterdam, The Netherlands: Elsevier, 1970, pp. 7–78.
- [70] E. van Wolde, *Words Become Worlds: Semantic Studies of Genesis 1–11*. Leiden, The Netherlands: Brill, 1994.
- [71] V. H. Yngve, "MT at the Massachusetts Institute of Technology," in *Proc. Nat. Symp. Machine Translation*, Los Angeles, H. P. Edmunson, Ed., Englewood Cliffs, NJ, USA: Prentice-Hall, 1961, pp. 126–132.
- [72] V. H. Yngve, "The COMIT system for mechanical translation," in *Proc. Inf. Process.: Proc. Int. Conf. Inf. Proc.*, Paris, France: UNESCO, 1960, pp. 183–187.
- [73] H. Zemanek, *Das Geistige Umfeld Der Informationstechnik*. Berlin, Germany: Springer, 1992.

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